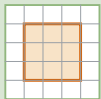


A Mammalian visual hierarchy



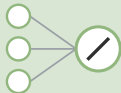
Local receptive fields

Each neuron samples a small region of the visual input



Simple cells

Detect oriented edges at specific positions



Complex cells

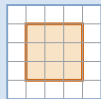
Aggregate nearby responses for position tolerance



Higher visual areas

Compose edges into shapes and recognizable objects

B Convolutional neural network



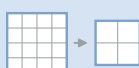
Local connectivity

Each kernel sees a small window of the image



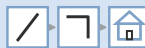
Convolution

Shared filters detect the same feature across positions



Pooling

Combines nearby responses while reducing spatial resolution



Stacked feature maps

Repeated local operations build edges into shapes and objects